Note

(i) If a matrix contains at-least one non-zero element, then $\rho(A) \geq 1$.

(ii) The rank of the identity matrix I_n is n.

(iii) If the rank of a matrix A is r, then there exists at-least one minor of A of order r which does not vanish and every minor of A of order $r+1$ and higher order (if any) vanishes.

(iv) If A is an $m \times n$ matrix, then $\rho(A) \leq \min\{m, n\}$ = minimum of m, n.

(v) A square matrix A of order n has inverse if and only if $\rho(A) = n$.

Example 1.15

Find the rank of each of the following matrices: (i) $\begin{bmatrix} 3 & 2 & 5 \\ 1 & 1 & 2 \\ 3 & 3 & 6 \end{bmatrix}$ (ii) $\begin{bmatrix} 4 & 3 & 1 & -2 \\ -3 & -1 & -2 & 4 \\ 6 & 7 & -1 & 2 \end{bmatrix}$

Solution

(i) Let $A = \begin{bmatrix} 3 & 2 & 5 \\ 1 & 1 & 2 \\ 3 & 3 & 6 \end{bmatrix}$. Then A is a matrix of order 3×3. So $\rho(A) \leq \min\{3, 3\} = 3$. The highest order of minors of A is 3. There is only one third order minor of A.

It is $\begin{vmatrix} 3 & 2 & 5 \\ 1 & 1 & 2 \\ 3 & 3 & 6 \end{vmatrix} = 3(6-6) - 2(6-6) + 5(3-3) = 0$. So, $\rho(A) < 3$.

Next consider the second-order minors of A.

We find that the second order minor $\begin{vmatrix} 3 & 2 \\ 1 & 1 \end{vmatrix} = 3 - 2 = 1 \neq 0$. So $\rho(A) = 2$.

(ii) Let $A = \begin{bmatrix} 4 & 3 & 1 & -2 \\ -3 & -1 & -2 & 4 \\ 6 & 7 & -1 & 2 \end{bmatrix}$. Then A is a matrix of order 3×4. So $\rho(A) \leq \min\{3, 4\} = 3$.

The highest order of minors of A is 3. We search for a non-zero third-order minor of A. But we find that all of them vanish. In fact, we have

$$\begin{vmatrix} 4 & 3 & 1 \\ -3 & -1 & -2 \\ 6 & 7 & -1 \end{vmatrix} = 0; \quad \begin{vmatrix} 4 & 3 & -2 \\ -3 & -1 & 4 \\ 6 & 7 & 2 \end{vmatrix} = 0; \quad \begin{vmatrix} 4 & 1 & -2 \\ -3 & -2 & 4 \\ 6 & -1 & 2 \end{vmatrix} = 0; \quad \begin{vmatrix} 3 & 1 & -2 \\ -1 & -2 & 4 \\ 7 & -1 & 2 \end{vmatrix} = 0.$$

So, $\rho(A) < 3$. Next, we search for a non-zero second-order minor of A.

We find that $\begin{vmatrix} 4 & 3 \\ -3 & -1 \end{vmatrix} = -4 + 9 = 5 \neq 0$. So, $\rho(A) = 2$.

Remark

Finding the rank of a matrix by searching a highest order non-vanishing minor is quite tedious when the order of the matrix is quite large. There is another easy method for finding the rank of a matrix even if the order of the matrix is quite high. This method is by computing the rank of an equivalent row-echelon form of the matrix. If a matrix is in row-echelon form, then all entries below the **leading diagonal** (it is the line joining the positions of the diagonal elements $a_{11}, a_{22}, a_{33}, \cdots$ of the matrix) are zeros. So, checking whether a minor is zero or not, is quite simple.

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